EC4L01 ANALOG CMOS IC - 2 LAB

DEPARTMENT OF ELECTRONICS AND COMMUNICATION,
NIE, MYSORE
EVEN Semester AY 2025-26

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Experiment 5

Aim:

Design and analyze various op-amp circuits: i) Zero crossing detector ii) Comparator iii) inverting iv) non-inverting amplifier and v) voltage follower configurations using uA741 IC

Simulation Tool: Tina TI

Components Required: µA741 IC(Op-Amp), resistors, signal generator, battery, DMM

Schematic Circuit:

A) Zero Crossing Detector

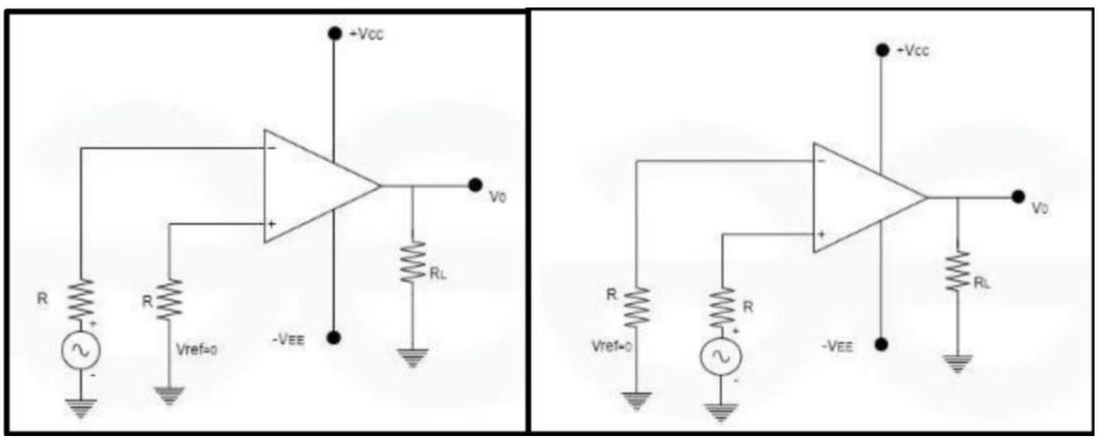
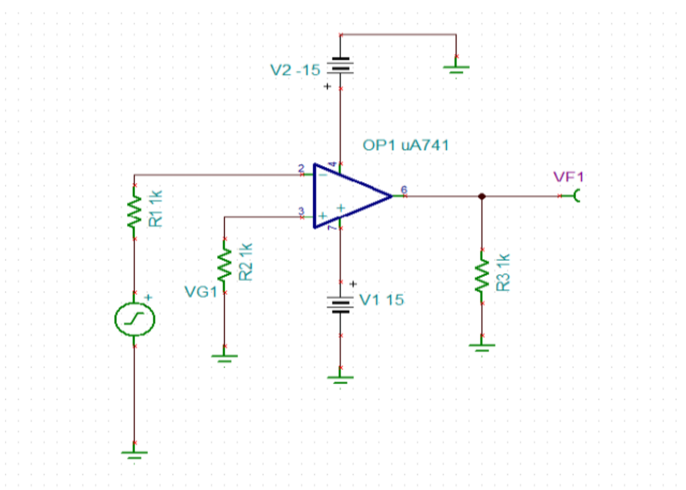


Fig. 1 a) Inverting ZCD b) Non-inverting ZCD

1a) Inverting ZCD
Design:



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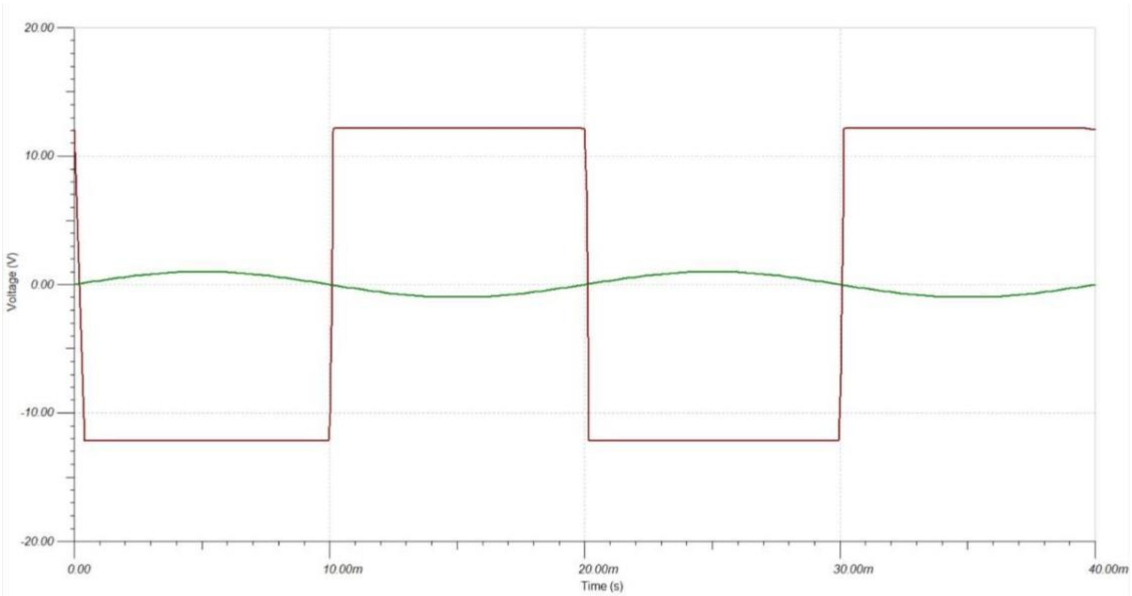
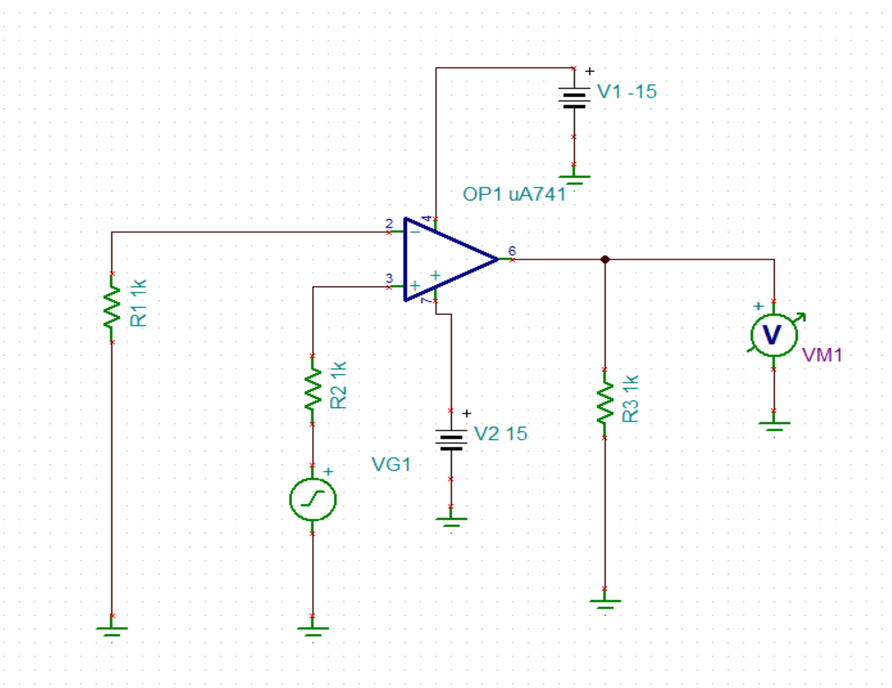


1b) Non Inverting ZCD Design



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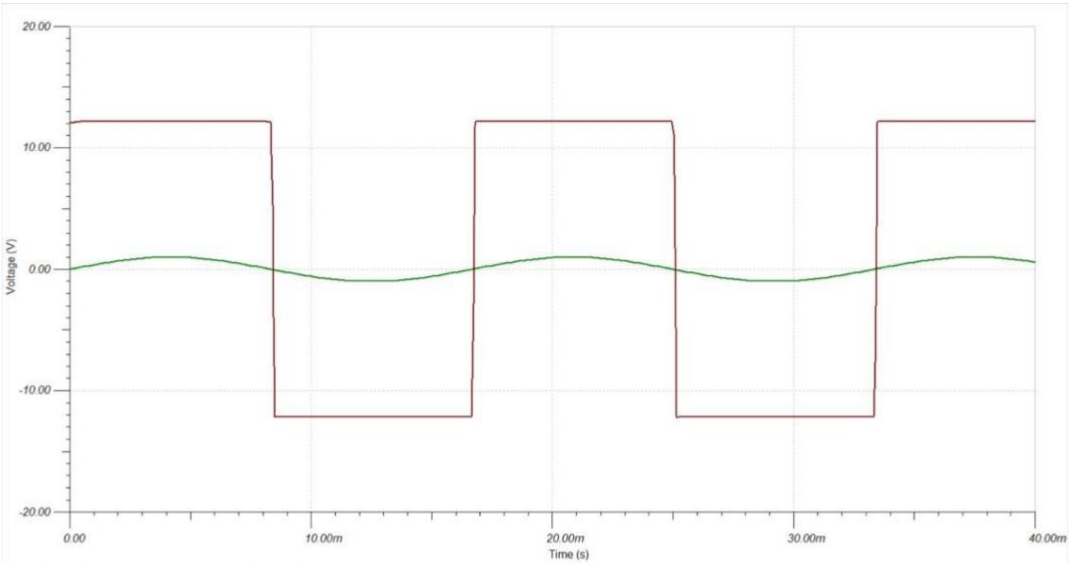
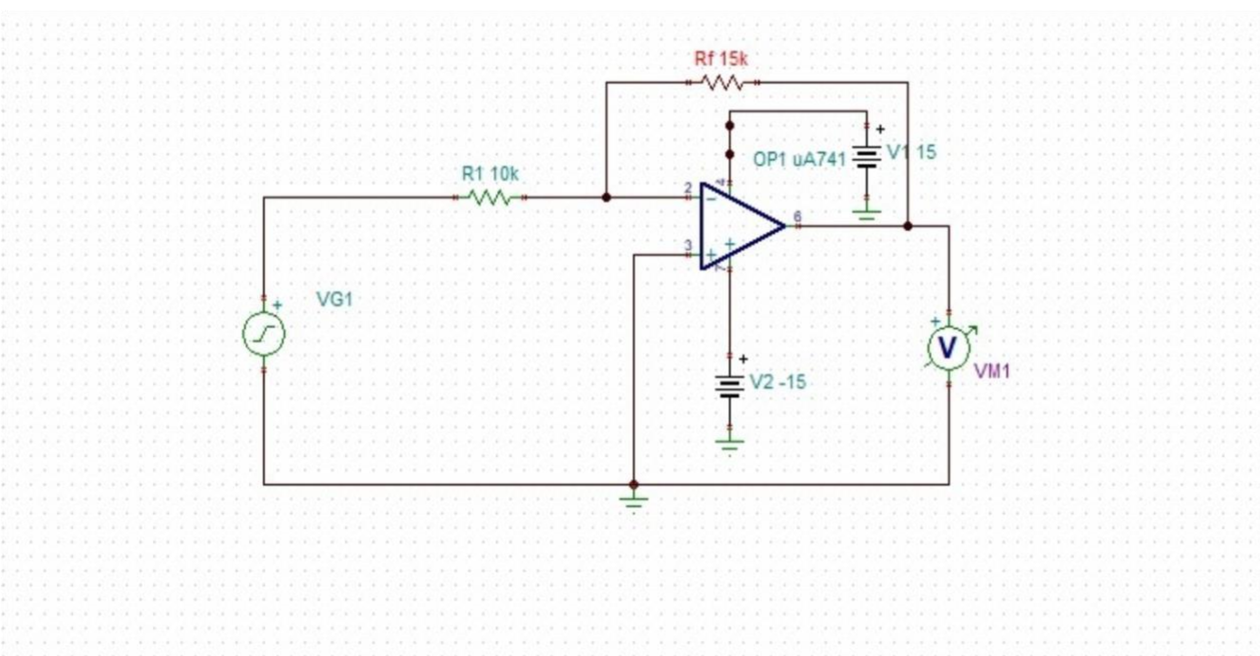


2a) Inverting Amplifier

Design: $V_{out} = -1.5V_{in}$. Assume R_f = 15k ohm R_1 = 10k ohm



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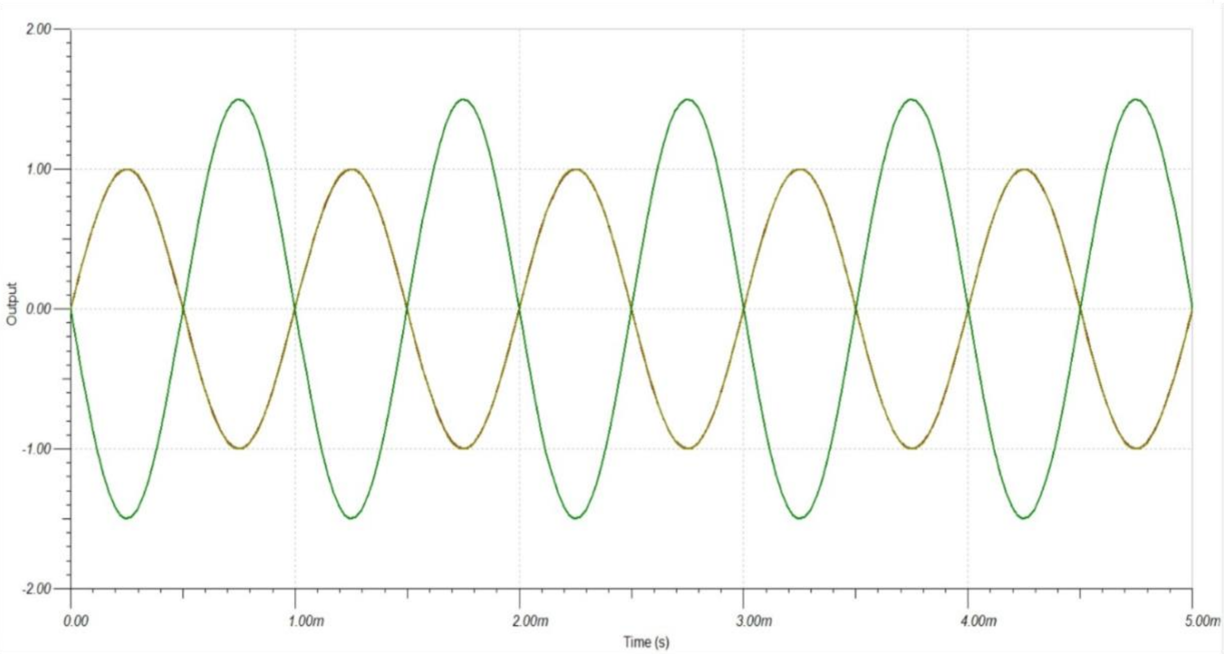
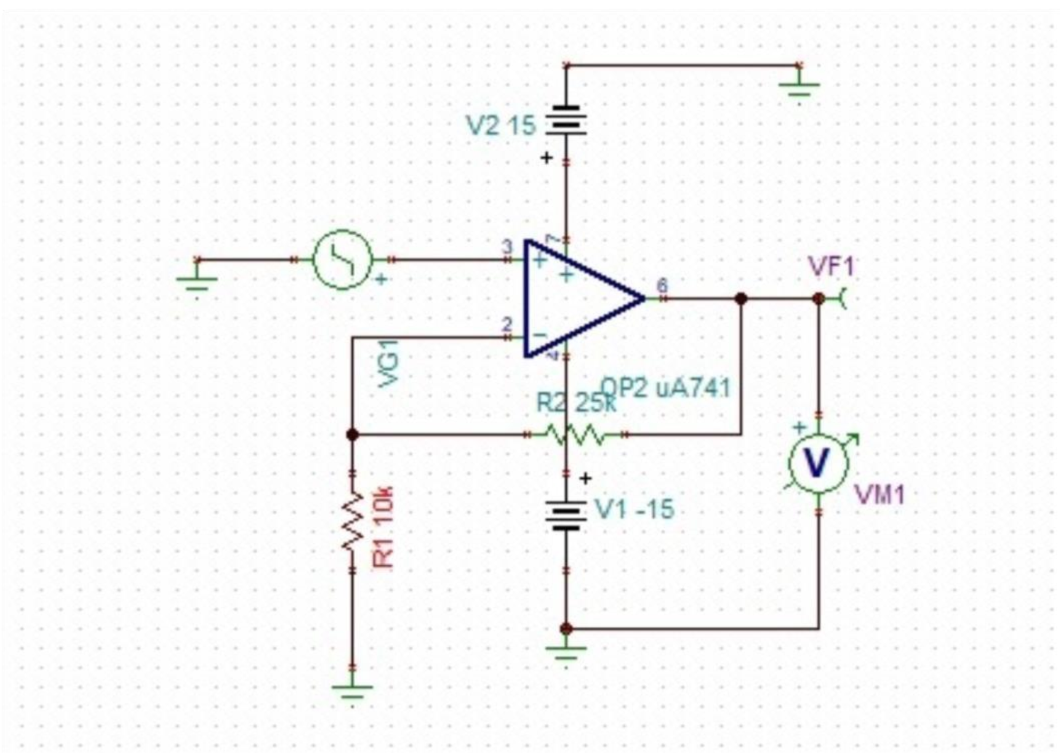


2b) Non Inverting Amplifier

Design: $V_{out} = 3.5V_{in}$. Assume R_f = 25k ohm R_1 = 10k ohm



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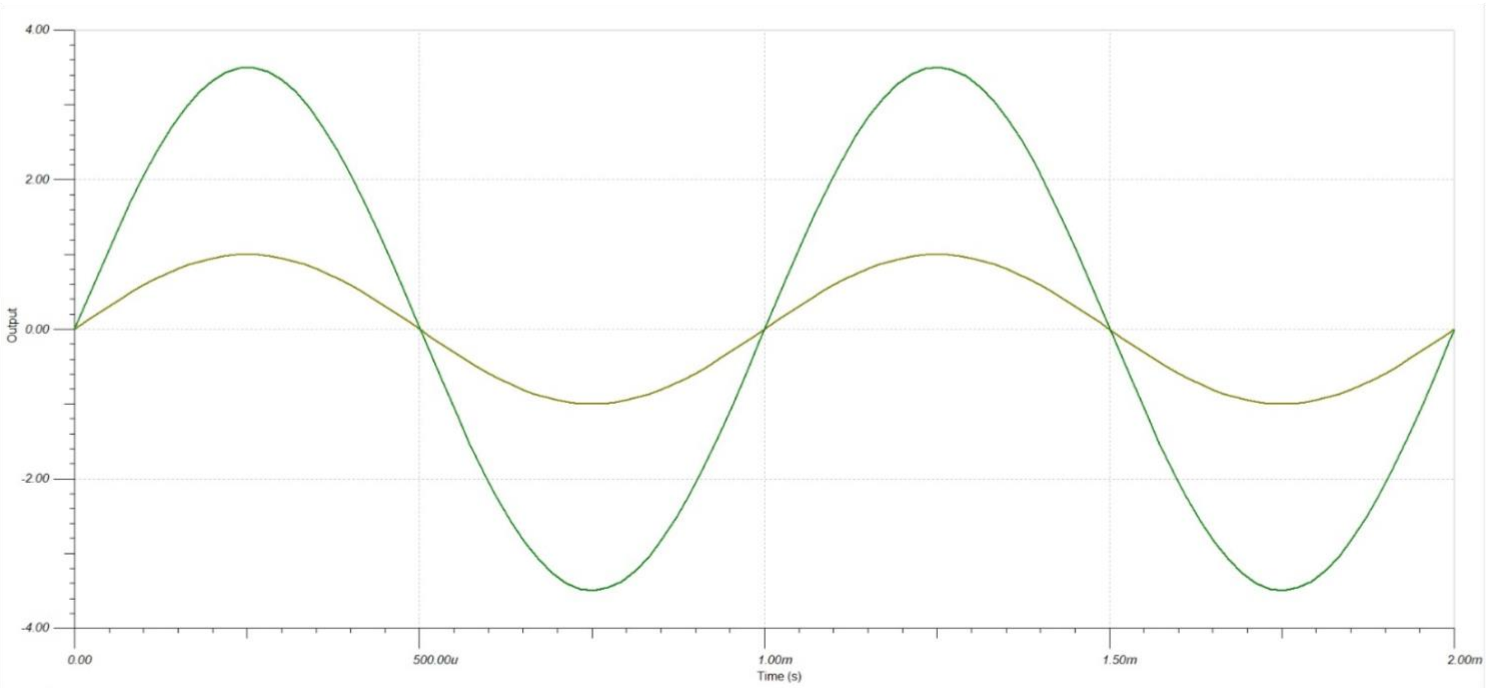
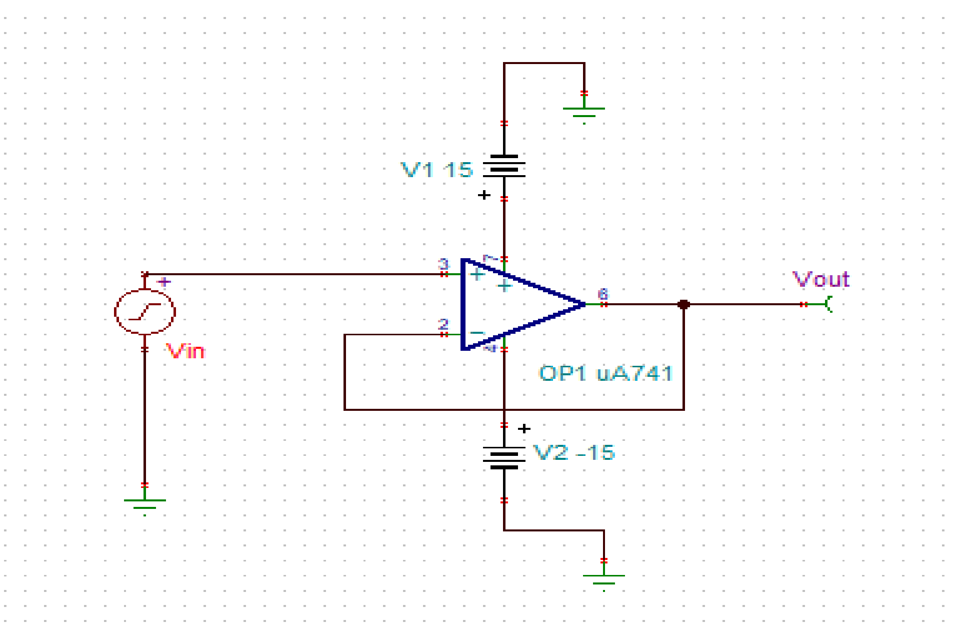
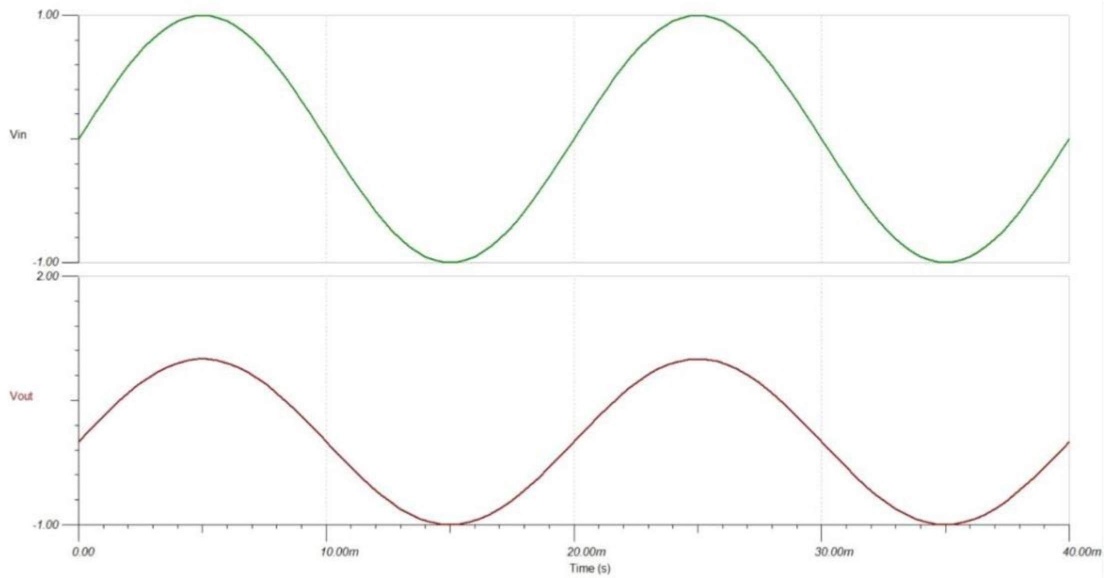


3) Voltage Follower

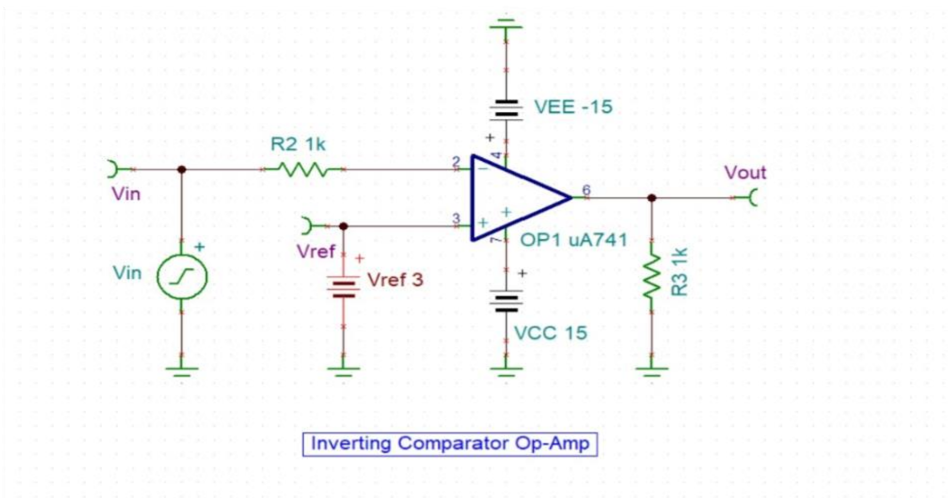
Design:





4) Inverting Comparator

Design: Vref=3v , Let's Assume Rf=1k ohm R1=R2=1k ohm.



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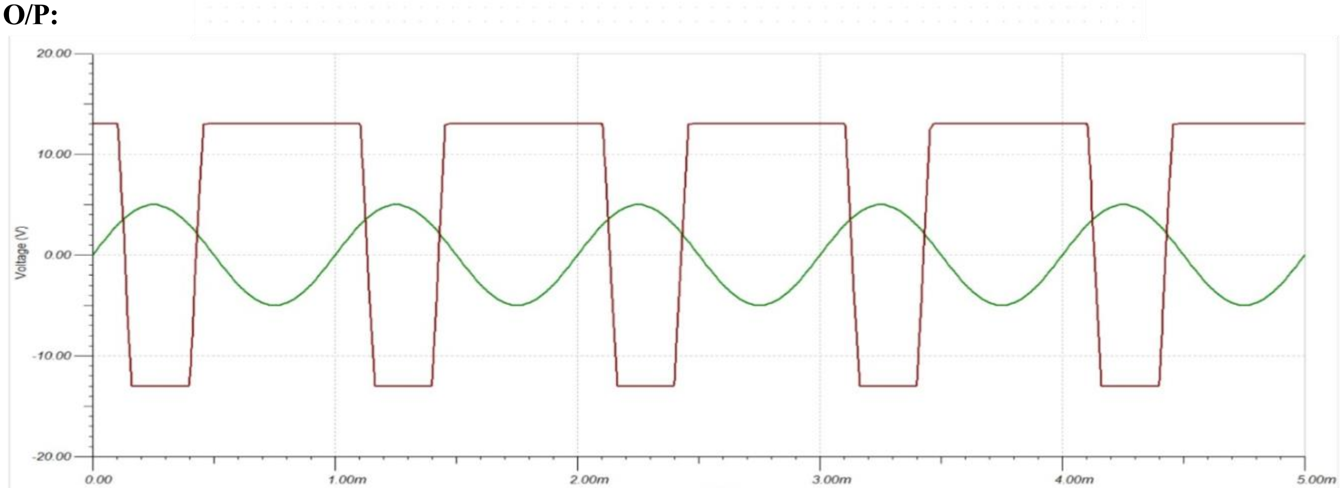
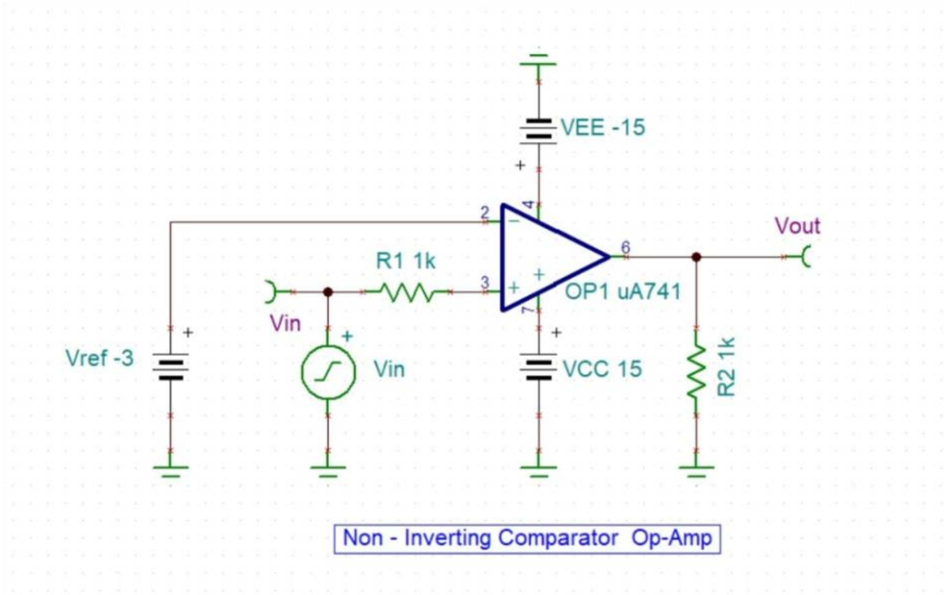


5) Non-inverting Comparator

Design:

Vref=-3v , Let's Assume Rf=1k ohm R1=R2=1k ohm.



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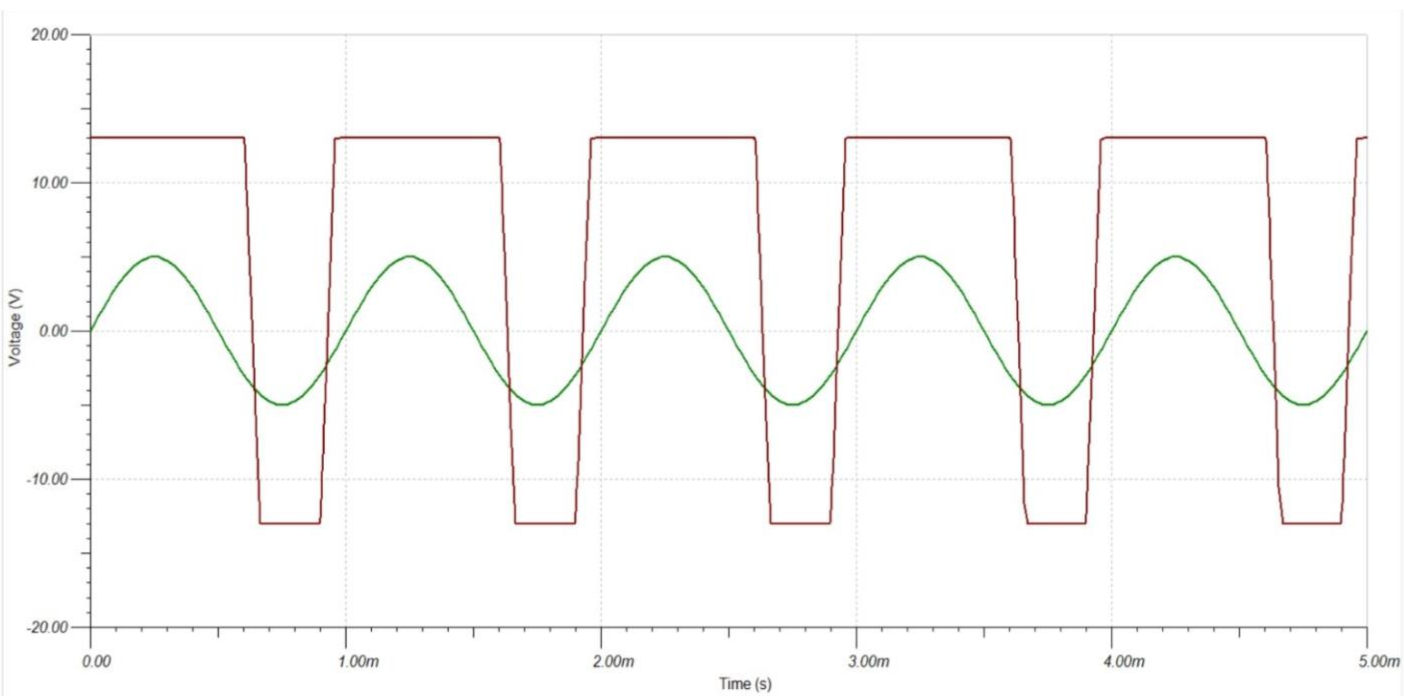

Inference:

1. The output switches exactly at the zero crossing of the input signal with sharp and nearly instantaneous transitions, confirming correct operation of the zero crossing detector, while saturating at approximately ±12–13 V instead of ±15 V due to practical limits of the µA741 model.
2. The output is properly amplified with the expected phase relationship (180° phase shift for inverting and same phase for non-inverting), showing correct circuit operation, while the gain is close to the designed value.
3. The output closely follows the input with the same phase and nearly equal amplitude, confirming unity gain operation of the voltage follower

Result:

1. Zero crossing detector, inverting amplifier, non-inverting amplifier, and voltage follower circuits using µA741 were successfully designed and simulated in TINA-TI.
2. The obtained output waveforms verify correct switching, amplification, and buffering operations as expected.

Verified by:

Date: